

Location Parameter MLE for Cauchy Distribution

Using Simulated Annealing

Problem: Find maximum likelihood estimate (MLE) of μ for Cauchy distribution with log-likelihood:

$$l(\mu) = -n \ln(\pi) - \sum_{i=1}^n \ln(1 + (X_i - \mu)^2)$$

Challenge: Log-likelihood is multimodal (non-concave); standard optimization methods fail.

Data: $X \sim t(df=1) + \mu^*$, where $\mu^* = 5$ (true location), $n = 4$ (sample size)

Algorithm: Simulated Annealing for Cauchy MLE

Input: N (iterations), r (step size), X (data)

Output: μ_{hat} (MLE estimate)

Step 1: Initialize

$$\mu_1 \leftarrow \text{Uniform}(-10, 40)$$

Step 2: For $k = 2$ to N :

(a) Generate candidate: $a \leftarrow \text{Uniform}(\mu_{k-1} - r, \mu_{k-1} + r)$

(b) Temperature schedule: $T_k \leftarrow 1/(1 + \ln(\ln(k)))$

(c) Evaluate likelihood ratio:
 $\alpha \leftarrow l(a)/l(\mu_{k-1}) = e^{l(a) - l(\mu_{k-1})}$

(d) Accept criterion:
if $u < \min(1, \alpha)$ where $u \sim \text{Uniform}(0,1)$:

$$\mu_k \leftarrow a$$

else:

$$\mu_k \leftarrow \mu_{k-1}$$

Step 3: Return $\mu_{\text{hat}} = \mu_k$ where $k = \arg \max \{l(\mu_i)\}$

Key Components:

1. Log-Likelihood Function:

$$l(\mu) = -n \ln(\pi) - \sum \ln(1 + (X_i - \mu)^2)$$

- Non-concave: multiple local maxima
- Algorithm avoids getting trapped in local maxima

2. Adaptive Temperature Schedule: $T_k = 1/(1 + \ln(\ln(k)))$

- Decreases faster than standard $1/\ln(k)$
- Suitable for discrete/bounded parameter spaces
- Early iterations: broad exploration of $[-10, 40]$
- Later iterations: convergence to optimum

3. Metropolis Acceptance: $\alpha = e^{l(a) - l(\mu_{k-1})}$

- If $\alpha > 1$: Accept always (better likelihood)
- If $\alpha \leq 1$: Accept with probability α
- Accepts worse solutions probabilistically

4. Step Size (Neighborhood): $a \leftarrow \text{Uniform}(\mu_{k-1} - r, \mu_{k-1} + r)$

- r controls exploration radius
- $r = 5$ (default): balanced exploration-exploitation
- $r = 500$ (large): many rejections, sparse sampling
- $r = 0.1$ (small): frequent moves, slower exploration

Results and Observations:

Convergence Behavior (Multiple Runs, $N=100$, $r=5$):

- Four independent runs all converge to same MLE estimate
- Each run explores multimodal likelihood surface
- Different starting values (μ_1 in $[-10, 40]$) converge to same optimum
- Sampled points form cloud around true likelihood peaks

Effect of Step Size r ($N=1000$):

- Large $r = 500$:
 - Many candidate rejections (proposals outside likelihood mass)
 - Sparse trajectory, poor exploration
 - Algorithm struggles with large neighborhood
- Small $r = 0.1$:
 - High acceptance rate (many small moves accepted)
 - Frequent moves but slow global exploration
 - May require more iterations to escape local optima

Convergence and Trade-offs:

- **Global Convergence:** Temperature schedule T_k goes to 0 ensures convergence to global MLE
- **Exploration-Exploitation Trade-off:**
 - High T_k early: Accept bad moves (explore multimodal surface)
 - Low T_k late: Reject bad moves (refine around optimum)
- **Step Size Optimization:** r must be tuned
 - Too large: rejection rate too high, inefficient
 - Too small: convergence slow, may miss global maximum
 - Optimal r is approximately parameter space spread divided by 10
- **MLE Property:** Returned estimate $\mu_{\text{hat}} = \arg \max \{l(\mu_i)\}$ is asymptotically normal under regularity conditions, even though initialization is random